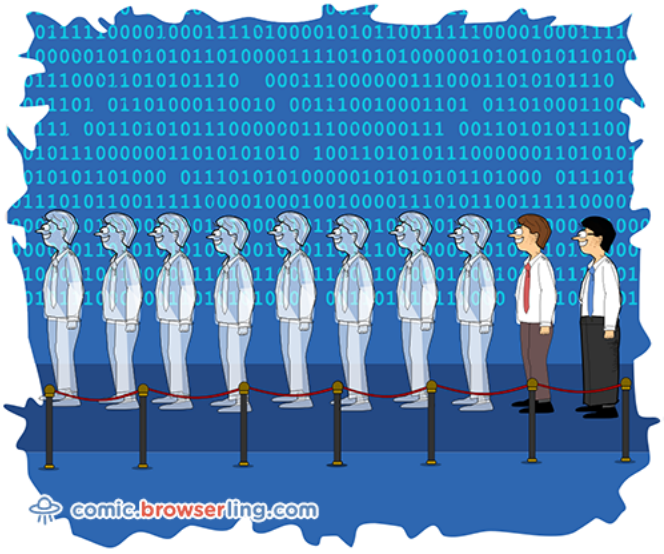
Homework #3: Computer Overview and Data Representation

CS 255: Computer Architecture and Machine Level Programming



There are only 10 types of people in the world:
Those who understand binary, and those who don't.

- To submit, complete the assignment on Gradescope.

1. Answer the following questions about the computer as defined in class.

   (a) Which components make up the logical structure of a computer?

   A. CPU True
   B. Memory True
   C. Input devices True
   D. Output devices True
   E. Power supply

   (b) Which of the following describes program instructions?

   A. Data that represents user input.
   B. Commands that tell the CPU what operations to perform. True
   C. Memory locations that store output.
   D. Hardware signals controlling the bus.

   (c) Why is the stored-program concept important?

   A. It allows programs to be permanently wired into the CPU.
   B. It enables changing program behavior by modifying memory contents. True
   C. It increases CPU clock speed.
   D. It removes the need for input devices.

   (d) When the CPU sends data to a display, which bus carries the actual bits of that data?

   A. Address bus
   B. Data bus True
   C. Control bus
   D. System bus

   (e) Why is binary representation used for all data types?

   A. The CPU only understands two voltage levels.True
   B. It simplifies programming languages.
   C. It uses less memory.
   D. It matches human arithmetic.

2. Answer the following questions about positional number systems.

   (a) Which statement best describes how all positional number systems work?

   A. Each digit's value is multiplied by a power of 10.
   B. Each digit's value depends on its position and the system's base. True
   C. Only binary and hexadecimal use positional values.
   D. The rightmost digit always represents one half.

   (b) Convert the unsigned binary number `101101` to decimal. 45

   (c) Convert the decimal number 27 to an unsigned binary number. 11011

   (d) Convert the hexadecimal number `0xabc43627` to binary. 10101011110001000011011000100111

   (e) Convert the base-3 number 10212 to decimal. 104

   (f) Which statement correctly describes the advantage of hexadecimal notation?

   A. It is easier for humans to read and write than long binary strings. True
   B. It allows the CPU to process data faster.
   C. It reduces memory usage.
   D. It encodes text more efficiently.

3. Answer the following questions about two's complement representation.

(a) Represent -7 as an 8-bit 2's-complement binary number. 11111001

(b) Interpret the 8-bit 2's-complement number 11110101 as a decimal integer. -11

(c) Evaluate 0101 − 0011 using binary arithmetic, assuming each number is a 4-bit 2's complement signed integer. 0010

(d) What is the numeric range of signed 8-bit 2's-complement integers? -128 to 127

(e) Using 4-bit 2's complement numbers, select all expressions that overflow?

A. 0101 + 0011 True
B. 0110 + 1001
C. 1100 + 1010 True
D. 1110 + 0001

(f) How do you obtain the 2's-complement representation of a binary number?

A. Flip all bits and add 1 True
B. Flip only the sign bit
C. Subtract 1 from the number
D. Add 1 to the sign bit

(g) Why does 2's-complement simplify arithmetic for the CPU?

A. It uses separate hardware for positive and negative numbers
B. The same addition circuitry works for signed and unsigned numbers True
C. It stores sign and magnitude separately
D. It avoids overflow conditions

4. Answer the following questions about floating point representation.

   (a) How is a 32-bit IEEE 754 single-precision floating-point number structured?

   A. 1 sign bit, 8 exponent bits, 23 fraction bits True
   B. 1 sign bit, 10 exponent bits, 21 fraction bits
   C. 8 sign bits, 8 exponent bits, 16 fraction bits
   D. 1 sign bit, 7 exponent bits, 24 fraction bits

   (b) What is the encoding scheme used for the exponent field in the IEEE 754 single-precision floating-point format?

   A. Two's complement
   B. Sign-magnitude
   C. Excess-127 True
   D. Unsigned integer

   (c) What decimal value is represented by the IEEE 754 single-precision bit pattern `01000000101000000000000000000000`? 5.0

   (d) Give the IEEE 754 single-precision representation of the decimal value -2.5 in hexadecimal format. 0xc0200000

5. Answer the following questions about ASCII representation.

   (a) The binary string `01000001` represents what ASCII character? A

   (b) What binary string corresponds to the ASCII character 'Z'? 01011010

   (c) What ASCII text does `01001000 01100101 01101100 01101100 01101111` represent? Hello